BHANU SURAJ MALLA – DATA SCIENTIST

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EDUCATION

Master's Business Analytics, **Arizona State University, Tempe, AZ**

Bachelor's Petroleum Engineering with Finance, **Indian Institute of Technology, INDIA**

RELEVANT WORK HISTORY

Amazon Inc., Seattle, WA **JUNE 2024 - Present**
Data Scientist II

***Mission**: Reduce hiring uncertainty for all of Amazon through probabilistic forecasting and scalable ML systems.*

- Built **attrition forecasting models** using Bayesian hierarchical state-space models, capturing volatile time-series behavior via MCMC to improve workforce planning accuracy by 25%. (PyMC, Bayesian Statistics)
- Designed **probabilistic hiring funnel metrics** (cycle times and conversion rates) across orgs, regions, and job families by estimating full posterior distributions rather than point estimates.
- Developed **medium-term hiring start forecasts** at multiple aggregation levels using an ensemble of LightGBM models incorporating historical starts, requisitions, and funnel signals. (scikit-learn)
- Built and maintained a **front-end data intake system** to capture recruiter productivity and headcount, enabling short-term start forecasts via attribution of open requisitions across org hierarchies. (Flask, JavaScript, HTML)
- Designed and deployed **GenAI system ("Automated Insights")** that converts structured data and prompts reducing manual analysis time by 50%. (Anthropic Claude)
- **Served as the team's infrastructure expert**, designing, and owning **core ML platform components** including Docker transforms, shared test packages, and deployment abstractions used across talent acquisition data science applications. (Typescript, AWS Cloud)
- **Built scalable CI/CD pipelines** that track package changes, automatically build, and publish Docker images, and provision containers for **AWS Lambda and Batch workloads**, enabling reliable, multi-region deployments **at Amazon-wide scale**.

Circle K Stores Inc., Tempe, AZ **JAN 2023- JUNE 2024**
Senior Data Scientist

***Mission**: Maximize revenue and profits across stores with data-driven pricing, promotions, and loyalty programs.*

- **Built generalized site-clustering models** (K-Means, Gaussian Mixture Models) for **500+ carwash locations across Europe**, leveraging engineered site, location, and sales features to enable scalable segmentation. *(Databricks, Azure)*
- **Developed parametric and non-parametric demand forecasting models** at the cluster level to optimize **supply chain and pricing decisions**, delivering **$4M in cost savings**. *(PySpark, SQL, Neural Networks)*
- **Led ML lifecycle and model governance frameworks** using **MLflow**, managing **1,000+ forecasting and clustering models** across production environments. *(MLOps, DevOps)*
- **Designed a time-series stationarity toolkit** implementing **Perron, KPSS, and ADF tests** to guide data transformations, improving forecasting accuracy (**W-MAPE ↓ 10%**).

Circle K Stores Inc., Tempe, AZ **JULY 2021-DEC 2022**
Data Scientist

- **Built a scalable measurement ETL pipeline on Databricks**, integrating **thousands of product, transaction, store, and cost datasets** to deliver automated revenue reporting and actionable insights for business stakeholders. *(SQL, Power BI, pandas, NumPy)*
- **Partnered with Canadian business units** to translate pricing constraints into a **non-linear optimization model** for localized pricing, generating **millions in quarterly revenue**. *(Python, R, Pyomo)*
- **Modeled price elasticity at the store–item level** using **unobserved components models (UCM)** and **log–log**

regression to estimate demand response to price changes. *(Python, R)*

- **Designed and executed A/B tests for pricing strategies**, driving a **7% margin lift** and **12% increase in units sold** across North American business units. *(Statistics)*

PROVIDERTRUST INC., NASHVILLE, TN **JUNE 2020 – JULY 2021**
ETL Specialist

Mission: *Maintain reliable healthcare compliance data pipelines and minimize ETL downtime.*

- **Built large-scale ETL pipelines** ingesting provider licensing, exclusion, and sanction data from **500+ U.S. state compliance board websites** using automated web extraction. *(Python, Selenium, JavaScript, Jenkins)*
- **Designed a GUI-driven web-scraping framework** that generates scrapers from templates and auto-populates page element signatures from user input, **cutting scraper development time by 50%**. *(Tkinter, Regex)*
- **Developed a reusable Python parsing library** to standardize and validate external vendor data, **reducing pipeline error rates by 7%** and improving downstream data quality. *(pandas, NumPy, Amazon Redshift)*
- **Implemented real-time ETL observability dashboards** to track downtime, root causes, and remediation signals, **reducing ETL downtime by 30%** through proactive monitoring. *(Tableau, AWS CloudWatch)*

CAIRN OIL AND GAS, NEW DELHI, INDIA **JULY 2017 - JULY 2019**
Engineer

Mission: *Maximize crude oil production while minimizing upstream costs through simulation and optimization.*

- **Researched and simulated a novel surface transportation approach** for oil and gas using **multiphase network models**, evaluating phase interactions to reduce transportation cost and time. *(Simulation, Fluid Dynamics)*
- **Built decline curve analysis models** to forecast well production using historical output from geologically similar wells, improving production planning accuracy. *(Statistics, Time Series)*
- **Automated inventory tracking and allocation** for **1,000+ assets across drilling projects** by integrating SAP data and spreadsheets, optimizing inventory sharing across rigs. *(Optimization)*
- **Designed a benchmarking analytics dashboard** analyzing **200+ drilling rigs over 30 years** to identify top-performing rigs and operational best practices. *(Python, SQL, Tableau)*

PYTHON PACKAGE PUBLICATIONS

Package Name: MIC (Make-it-Certain)
Publication: https://make-it-certain.streamlit.app/
GitHub: https://**github.com**/mbsuraj/MIC

Brief Description: MIC is designed to provide certainty in time series predictions by comparing multiple forecasting approaches across various datasets. The framework automatically handles data preprocessing, model training, hyperparameter optimization, and performance evaluation. Also has a user-friendly web interface that allows non-technical users to run forecasting experiments interactively.

Package Name: StationarityToolkit
Publication: https://**pypi.org**/project/stationarityToolkit/
GitHub: https://**github.com**/mbsuraj/stationarityToolkit

Brief Description: This toolkit can make statistical inferences of a time-series to determine trend and variance non-stationarity and make appropriate transformations along with returning the inverse methods to these transformations.

TECHNOLOGY SKILLS & CERTIFICATIONS

Amazon Web Services	FastAPI	Research Writing	SQL
Azure	Docker	LLMs	Tableau
Databricks	Bash	R	PowerBI
Sagemaker	PySpark	Java	QuickSight
Jenkins	Git	Jira, Asana	Looker